

Practice Exam Set C — Solutions

Part I: Generative Models

Exercise A — Probabilistic PCA

Answer: A

Using the marginalisation formula $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \mathbf{b}, \mathbf{w}\mathbf{w}^\top + \mathbf{I})$ with $\mathbf{w} = [2, 1]^\top$ and $\mathbf{b} = [0, 1]^\top$:

$$\mathbf{w}\mathbf{w}^\top = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}, \quad \mathbf{w}\mathbf{w}^\top + \mathbf{I} = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}.$$

$$p(\mathbf{x}) = \mathcal{N}\left(\mathbf{x} \mid [0, 1]^\top, \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}\right).$$

Exercise B — Normalizing flows

Answer: B

$T(\mathbf{u}) = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{u} + [1, -1]^\top$, so $\det J_T = 3 \cdot 2 = 6$ and $T^{-1}(\mathbf{x}) = \frac{1}{6} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} (\mathbf{x} - [1, -1]^\top)$.

At $\mathbf{x} = [1, -1]^\top$: $T^{-1}([1, -1]^\top) = [0, 0]^\top$. $p_{\mathbf{u}}([0, 0]^\top) = \frac{1}{2\pi} e^0 = \frac{1}{2\pi}$.

$$p_{\mathbf{x}}([1, -1]^\top) = \frac{p_{\mathbf{u}}([0, 0]^\top)}{|\det J_T|} = \frac{1/2\pi}{6} = \boxed{\frac{1}{12\pi}}.$$

Exercise C — DDPM forward posterior

First compute $\bar{\alpha}_1$ and $\bar{\alpha}_2$:

$$\bar{\alpha}_1 = 1 - \beta_1 = 1 - \frac{1}{4} = \frac{3}{4}, \quad \bar{\alpha}_2 = (1 - \beta_1)(1 - \beta_2) = \frac{3}{4} \cdot \frac{2}{3} = \frac{1}{2}.$$

Apply the forward posterior formula at $t = 2$ (so $t - 1 = 1$), $\mathbf{x}_0 = [0, 0]^\top$, $\mathbf{x}_2 = [3, 0]^\top$:

$$\tilde{\beta}_1 = \frac{\beta_2(1 - \bar{\alpha}_1)}{1 - \bar{\alpha}_2} = \frac{\frac{1}{3} \cdot \frac{1}{4}}{\frac{1}{2}} = \frac{1}{6}.$$

$$\tilde{\mu}_1 = \frac{\sqrt{\bar{\alpha}_1} \beta_2}{1 - \bar{\alpha}_2} \mathbf{x}_0 + \frac{\sqrt{1 - \beta_2}(1 - \bar{\alpha}_1)}{1 - \bar{\alpha}_2} \mathbf{x}_2.$$

Since $\mathbf{x}_0 = [0, 0]^\top$, the first term vanishes:

$$\tilde{\mu}_1 = \frac{\sqrt{2/3} \cdot \frac{1}{4}}{\frac{1}{2}} [3, 0]^\top = \frac{\sqrt{2/3}}{2} [3, 0]^\top = \frac{3\sqrt{2/3}}{2} [1, 0]^\top = \frac{\sqrt{6}}{2} [1, 0]^\top.$$

$$q(\mathbf{x}_1 \mid \mathbf{x}_2 = [3, 0]^\top, \mathbf{x}_0 = [0, 0]^\top) = \mathcal{N}\left(\mathbf{x}_1 \mid \left[\frac{\sqrt{6}}{2}, 0\right]^\top, \frac{1}{6}\mathbf{I}\right).$$

Part II: Representation Learning

Exercise D — Pull-back metric

D.1: For $f(\mathbf{x}) = [x_1^2 + x_2, x_1 + x_2^2, x_1x_2]^\top$:

$$J_{\mathbf{x}} = \begin{bmatrix} 2x_1 & 1 \\ 1 & 2x_2 \\ x_2 & x_1 \end{bmatrix}.$$

$$G_{\mathbf{x}} = J^\top J = \begin{bmatrix} (2x_1)^2 + 1 + x_2^2 & 2x_1 + 2x_2 + x_1x_2 \\ 2x_1 + 2x_2 + x_1x_2 & 1 + (2x_2)^2 + x_1^2 \end{bmatrix} = \begin{bmatrix} 4x_1^2 + x_2^2 + 1 & 2x_1 + 2x_2 + x_1x_2 \\ 2x_1 + 2x_2 + x_1x_2 & x_1^2 + 4x_2^2 + 1 \end{bmatrix}.$$

D.2: At $\mathbf{x} = [1, 0]^\top$:

$$G|_{[1,0]^\top} = \begin{bmatrix} 4 + 0 + 1 & 2 + 0 + 0 \\ 2 + 0 + 0 & 1 + 0 + 1 \end{bmatrix} = \boxed{\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}}.$$

Exercise E — Geodesic approximation

E.1

Answer: B

The objective $\sum_n \|f(\mathbf{c}_n) - f(\mathbf{c}_{n-1})\|$ is the total Euclidean length of the piecewise-linear path in *observation space* $\mathbb{R}^D$, which lies on or near the manifold $f(\mathbb{R}^d)$. Minimising it finds the shortest path **on the manifold**.

E.2

Answer: B

Minimising $\sum_n \|\mathbf{c}_n - \mathbf{c}_{n-1}\|$ gives a straight line in the latent space, not a geodesic on the manifold. Since f is nonlinear it generally distorts distances, so the two objectives disagree.

Exercise F — Curve energy

F.1: For $f(\mathbf{x}) = [x_1, x_1^2 + x_2^2, x_2]^\top$:

$$J_{\mathbf{x}} = \begin{bmatrix} 1 & 0 \\ 2x_1 & 2x_2 \\ 0 & 1 \end{bmatrix}, \quad G_{\mathbf{x}} = J^\top J = \begin{bmatrix} 1 + 4x_1^2 & 4x_1x_2 \\ 4x_1x_2 & 4x_2^2 + 1 \end{bmatrix}.$$

F.2: For $\mathbf{c}_1(t) = [t, 0]^\top$: $G_{\mathbf{c}_1(t)} = \begin{bmatrix} 1+4t^2 & 0 \\ 0 & 1 \end{bmatrix}$, $\dot{\mathbf{c}}_1^\top G \dot{\mathbf{c}}_1 = 1 + 4t^2$:

$$E(\mathbf{c}_1) = \int_0^1 (1 + 4t^2) dt = 1 + \frac{4}{3} = \boxed{\frac{7}{3}}.$$

For $\mathbf{c}_2(t) = [0, t]^\top$: $G_{\mathbf{c}_2(t)} = \begin{bmatrix} 1 & 0 \\ 0 & 4t^2 + 1 \end{bmatrix}$, $\dot{\mathbf{c}}_2^\top G \dot{\mathbf{c}}_2 = 4t^2 + 1$:

$$E(\mathbf{c}_2) = \int_0^1 (4t^2 + 1) dt = \frac{4}{3} + 1 = \boxed{\frac{7}{3}}.$$

By symmetry of f in x_1 and x_2 , $E(\mathbf{c}_1) = E(\mathbf{c}_2)$.

Exercise G — Neural network reparametrization

G.1: We need $\alpha w_1 \text{ReLU}(w_4 x) = w_1 \text{ReLU}(w_2 x)$ for all x . For $x > 0$ (where ReLU is active): $\alpha w_1 w_4 x = w_1 w_2 x \Rightarrow w_4 = w_2/\alpha$. For $x \leq 0$: both sides are 0 provided $w_4 = w_2/\alpha$ and $\alpha > 0$.

$$\boxed{w_4 = w_2/\alpha.}$$

G.2: The map $(\alpha w_1, w_2/\alpha) \mapsto f$ gives the same function for every $\alpha > 0$. Thus the parameters (w_1, w_2) are **not identifiable**: observations can only recover the product $w_1 w_2$, not the individual factors.

G.3: With $\sigma(x) = x$ (linear): $f(x|w_1, w_2) = w_1 w_2 x$. The same non-identifiability persists — only the product $w_1 w_2$ is identifiable. The issue is the *scaling symmetry*, which arises for any activation that is homogeneous (including the identity).

Part III: Graph Models

Exercise H — Shallow embeddings

H.1

Answer: A

$P(A_{uv} = 1) = 0.5 \Leftrightarrow \sigma(\mathbf{z}_u^\top \mathbf{z}_v) = 0.5 \Leftrightarrow \mathbf{z}_u^\top \mathbf{z}_v = 0$. With $\mathbf{z}_u = [1, 0]^\top$: $\mathbf{z}_u^\top \mathbf{z}_v = x = 0$. So $\mathbf{z}_v$ lies on the line $x = 0$ (the y -axis).

H.2

Answer: A

$\mathbf{z}_u^\top \mathbf{z}_u = 4 \Rightarrow P(A_{uu} = 1) = \sigma(4) \approx 0.982$. $\mathbf{z}_u^\top \mathbf{z}_v = 0 \Rightarrow P(A_{uv} = 1) = \sigma(0) = 0.5$. The self-loop (u, u) has higher probability.

Exercise I — Graph Laplacians and random walks

I.1: From L_{rw} : row a shows $-D^{-1}A_{ab} = -\frac{1}{3}$ for all $b \in \{b, c, d\}$, so node a has degree 3 with neighbours b, c, d . Rows b, c, d each show $-D^{-1}A_{?a} = -1$ (degree 1, sole neighbour a).

Sketch: star graph, centre a , leaves b, c, d .

I.2: Starting at b :

- Step 1: $b \rightarrow a$ with prob 1 (only neighbour).
- Step 2: $a \rightarrow b$ with prob $\frac{1}{3}$ (one of three leaves).

$$p = P(\text{at } b \text{ after 2 steps}) = \boxed{\frac{1}{3}}.$$

I.3: Stationary distribution $\pi_v \propto d_v$: $d_a = 3, d_b = d_c = d_d = 1$.

$$\boldsymbol{\pi} \propto [3, 1, 1, 1]^\top, \quad \text{normalised: } \boldsymbol{\pi} = \frac{1}{6}[3, 1, 1, 1]^\top.$$

Exercise J — Graph convolution (star graph)

J.1: $h[0] = 1, h[1] = -1, \mathbf{x} = [0, 1, 1, 1]^\top, A = \text{star adjacency (centre } a)$.

$$A\mathbf{x}: \text{row } a: 0 + 1 + 1 + 1 = 3; \quad \text{rows } b, c, d: 0.$$

$$A\mathbf{x} = [3, 0, 0, 0]^\top, \quad \mathbf{y} = \mathbf{x} - A\mathbf{x} = [0, 1, 1, 1]^\top - [3, 0, 0, 0]^\top = \boxed{[-3, 1, 1, 1]^\top}.$$

J.2: $M_{11} = h[0]\lambda_1^0 + h[1]\lambda_1^1 = 1 + (-1)(-\sqrt{3}) = 1 + \sqrt{3}$.

$$\boxed{M_{11} = 1 + \sqrt{3}}.$$